TROY PROGRAM	School Department Approval Assessment Session: **Final Exam** Form: **Mixed Exam** Duration: **60 minutes** Course ID: **ECO 251** Course name: **Principles of Macroeconomics** Code: **E14** Class ID: Exam Group: **Troy program** Term: **Spring 2024** Academic year: **2023–2024**

Student Name: _____	Student ID: _____	Signature: _____
Grade	**Examiners' Signature**	**Proctor's Signature**

PART 1: MULTIPLE-CHOICE QUESTIONS (50%)

Answer the multiple-choice questions in the table below by marking (x) in ONE corresponding box to the order of question, and answer options in the test.

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B															
C															
D															
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Q 1: First Bank of Country has $100,000 in checkable deposits and $100,000 in cash in its vault. The required reserve ratio in Country is 20%. What is the maximum amount of money that First Bank of Country can lend?
A. $20,000 B. $120,000 C. $100,000 D. $80,000

Q 2: The primary role of the Central Bank is to
A. Conduct monetary policy B. Respond to the requests of the Prime Minister
C. Print currency D. Oversee bank regulation

Q 3: Wing is a small island nation. Suppose that producers in Wing experience a devastating earthquake, drought, and plague of locusts that decrease productivity for many years. How will this affect the short-run aggregate supply (SRAS) curve and the long-run aggregate supply (LRAS) curve?
A. SRAS decreases; LRAS increases B. SRAS increases; LRAS increases
C. SRAS decreases; LRAS decreases D. SRAS increases; LRAS decreases

Q 4: A possible benefit from inflation is:
A. If nominal wages are fixed, inflation decreases real wages.
B. Inflation reduces distortions to relative prices.
C. If nominal wages are fixed, inflation increases real wages.
D. Inflation causes restaurants to update their menus more often.

Q 5: To obtain national income, start with GNP and subtract:
A. Depreciation, indirect business taxes, corporate profits, and social insurance contributions
B. Depreciation
C. Depreciation, indirect business taxes, and corporate profits
D. Depreciation and the statistical discrepancy

Q 6: A general increase in the price level is called:
A. Devaluation B. Inflation C. Hyperinflation D. Deflation

Q 7: Economists explain the downward-sloping aggregate demand curve with the interest rate effect. The idea is that lower price levels induce more savings, lower interest rates, and
A. Higher levels of import spending B. Lower levels of consumption spending
C. Lower levels of government spending D. Higher levels of investment spending

Q 8: The investment function and the IS curve slope:
A. Upward because higher interest rates induce less investment. B. Downward because higher interest rates induce less investment.
C. Upward because higher interest rates induce more investment. D. Downward because higher interest rates induce more investment.

Q 9: In the Keynesian cross model, if the interest rate is constant and the MPC is 0.7, then the government purchases multiplier is:
A. 0.3 B. 3.3 C. 0.7 D. 1.4

Q 10: The government is running a budget deficit if:
A. Tax revenue is greater than consumption spending. B. Tax revenue is greater than government spending.
C. Government spending is greater than tax revenue. D. Tax revenue is greater than investment spending.

Q 11: The unemployment resulting from wage rigidity and job rationing is called:
A. The natural rate of unemployment. B. Structural unemployment.
C. Insiders versus outsiders. D. The discouraged-worker effect.

Q 12: What would cause aggregate demand curve shift to the right?
A. A reduction in consumption spending B. An increase in import spending
C. An increase in the price level D. An increase in government spending

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Q 13: All other things equal, if the price of foreign-made cars rises, then the GDP deflator:
A. Will rise and the CPI will remain the same.
B. And the CPI will rise by different amounts.
C. Will remain the same and the CPI will rise.
D. And the CPI will rise by equal amounts.

Q 14: If the Central Bank increased the real amount of money supply, then the LM curve would:
A. Become steeper.
B. Become flatter.
C. Shift downward.
D. Shift upward.

Q 15: Monetary policy is the use of ___ to affect macroeconomic variables.
A. The money supply
B. Government spending
C. Real estate
D. Taxes

Q 16: All of the following is considered investment spending, EXCEPT:
A. Financial investment.
B. Inventory investment.
C. Business fixed investment.
D. Residential investment.

Q 17: Suppose that a farmer grows wheat and sells it to a baker for $1, the baker makes bread and sells it to a store for $2, and the store sells it to the customer for $3. This series of transactions increases GDP by:
A. $6.
B. $2.
C. $3.
D. $1.

Q 18: The difference between the nominal interest rate and the real interest rate is:
A. Hyperinflation.
B. Taxes.
C. Seigniorage.
D. The inflation rate.

Q 19: The unemployment caused by the time that it takes to match workers and jobs is called:
A. The discouraged-worker effect.
B. Frictional unemployment.
C. Wage rigidity.
D. Structural unemployment.

Q 20: The short-run aggregate supply curve is ___ because it is believed that ___ do not change quickly in the short run.
A. Upward sloping; wages
B. Upward sloping; interest rates
C. Vertical; wages and prices
D. Upward sloping; real output levels

Q 21: GDP measures:
A. Total income of everyone in the economy.
B. Total value added by all firms in the economy.
C. Expenditure on all final goods and services.
D. All of the answers are correct.

Q 22: Private saving is equal to:
A. Income - consumption.
B. Income - consumption - government spending.
C. Income - consumption - government spending - taxes.
D. Income - consumption - taxes.

Q 23: The relationship between the interest rate and the level of income that equilibrates the money market is called the:
A. LM curve.
B. Aggregate supply curve.
C. Aggregate demand curve.
D. IS curve.

Q 24: Which of the following events will cause the unemployment rate to increase?
A. An increase in the labor force with no change in the number of employed workers
B. A proportionally equal increase in the labor force and the number of unemployed workers
C. An increase in the number of employed workers with no change in the number of unemployed workers
D. An increase in population, with no change in the size of the labor force

Q 25: The IS curve is drawn for a given:
A. Monetary policy.
B. Fiscal policy.
C. Interest rate.
D. Level of income.

Q 26: According to Keynes, consumption is primarily determined by:
A. The interest rate.
B. Income and the interest rate.
C. Income.
D. The savings rate.

Q 27: Which of the following is NOT a social cost of inflation?
A. People hold smaller real balances and so have to make more frequent trips to the bank.
B. Inflation leads to greater variability in the relative prices charged by firms.
C. The money that people hold loses value due to the inflation tax.
D. Firms have to spend money to change prices more frequently.

Q 28: In the Keynesian cross model, if the interest rate is constant, the MPC is 0.6, and taxes are increased by $100, by how much does income change?
A. It decreases by $166.
B. It decreases by $150.
C. It increases by $166.
D. It increases by $150.

Q 29: Which of the following is NOT a policy tool that the Central Bank has used?
A. Changing the discount rate
B. Changing the reserve requirement
C. Conducting open market operations
D. Changing tax rates

Q 30: Which of the following policies would reduce the amount of frictional unemployment?
A. An increase in the minimum wage
B. A reduction in corporate taxes
C. An increase in unemployment insurance
D. Public retraining programs

Part 2: Problems Solving (50%)

Problem 1: (30%)

The data of goods market in an open economy as follows:

$C_o = 500;\ I_o = 400;\ G_o = 450;\ T_o = 450;\ X_o = 300;$ mpc = 0.7; mpm = 0.1

$AE = C + I + G + (X - M)$

A. Construct the planned aggregate expenditure function (AE/Yad) and find the equilibrium output/income? Illustrate the equilibria by your graph.

B. Compute the value of open market multiplier?

C. Find the net export (NX) at equilibrium output?

D. How much government spending should be changed (ΔG) to bring the economy to potential output of 3450?

Problem 2: (20%)

Using the following data:

Year	2020	2021	2022	2023
Nominal GDP (billion dong)	43,000	45,700	48,900	52,000
GDP Deflator (%)	100	103	107	110

A. Calculate real GDP of 2020, 2021, 2022 and 2023?

B. Compute economic growth rate in 2023 compared to 2022 and in 2021 compared to 2020?

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